Theory Assignment:- Solutions

① Explain the concept of a vector in R^n. How does the mathematical definition of a vector relate to feature representation in machine learning?

Ans:- A vector in R^n is an ordered n-tuple of real numbers written as

$$x = (x_1, x_2, \ldots, x_n)^T$$

each component represent a coordinate along a specific axis in n-dimensional space. In machine learning, each coordinate corresponds to a feature of a data point. Thus, a Dataset with n features is mathematically represented as vectors in R^n.

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② Define a vector space over R. Discuss why closure under addition and scalar multiplication are essential properties.

Ans:- A vector space over R is a set V equipped with vector addition and scalar multiplication such that for all $u, v \in V$ and $c \in R$.

$$u+v \in V, \quad cu \in V$$

closure under addition ensures combing two elements remains within the space.

closure under scalar multiplication: ensures scaling a vector does not leave the space. These properties gurantee algebraic consistency.

③ What is a subspace? State and explain the three necessary conditions required to prove that a subset is a subspace?

Ans:- A subspace H of a vector space V is a subset that is itself a vector space. It must satisfy:

ⓐ The zero vector $0 \in H$

ⓑ Closure under addition: $u, v \in H$, then $u+v \in H$

ⓒ Closure under scalar multiplication: if $u \in H$ and $c \in R$, then $cu \in H$.

④ Explain the concept of linear combination. why is linear combination fundamental in understanding span and basis?

Ans:- A linear combination of vectors $v_1, \ldots, v_p$ is $c_1v_1 + c_2v_2 + \cdots + c_pv_p$

linear combinations describe all possible weighted sums of vectors and form the foundation of span and basis.

⑤ Define the span of a set of vectors. Explain it's geometric meaning in R^2 and R^3.

Ans:- The span of vectors $\{v_1, \ldots, v_p\}$ is the set of all linear combinations of those vectors: $\text{span}\{v_1, \ldots, v_p\}$

In R^2, the span of one non-zero vector forms a line through the origin. Two linearly independent vectors span the entire plane. In R^3, one vector spans a line, two independent vectors span a plane, and three independent vectors span the entire space.

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⑤ What is linear independence? Explain the difference between linear independence and linear dependence with a clear mathematical condition.

Ans:- A set of vectors is linearly independent if $c_1v_1 + \cdots + c_pv_p = 0$ implies $c_1 = \cdots = c_p = 0$. If there exists a non trivial solution (some $c_i \neq 0$), the vectors are linearly dependent.

⑦ Discuss the problem of redundant features in machine learning from the perspective of linear dependence.

Ans:-

In machine learning, redundant features occur when one feature can be expressed as a linear combination of others. This creates linear dependence, leading to multicollinearity, unstable parameter estimates, and reduced model interpretability.

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⑧ Define a basis of a vector space? why must a basis satisfy both spanning and linear independence conditions?

Ans:

A basis of a vector space V is a set of vector that:

ⓐ Span V

ⓑ Are linearly independent

Spanning ensures coverage of the entire space, while independence ensure no redundancy.

(9) What is the dimension of a vector space? Explain its relationship with basis and matrix rank.

Ans:- The dimension of a vector space is the number of vectors in any basis of the space.

$$\dim(V) = \text{number of basis vectors}$$

It is equal to the Rank of a matrix whose rows (or column) form a basis, since rank represents the Number of linearly independent vectors.

(10) Explain the importance of understanding vectors space in machine learning, especially in dimensionality reduction techniques such as PCA.

Ans:- Understanding vector space allow us to analyze data geometrically. Dimensionality reduction method such as PCA identify a lower-dimensional subspace that captures maximum variance. This improves computational efficiency and reduce redundancy while preserving essential structure.

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